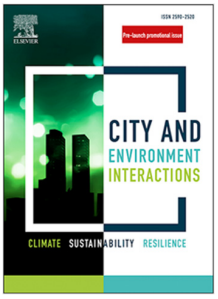
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Predicting environmental health-related search patterns using freely available digital and reanalysis data: A critical evaluation of google trends and global datasets for Morocco

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Predicting Environmental Health-Related Search Patterns Using Freely Available Digital and Reanalysis Data: A Critical Evaluation of Google Trends and Global Datasets for Morocco

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ABSTRACT

Climate change and environmental degradation increasingly threaten urban public health, yet monitoring infrastructure is often unavailable in resource-constrained settings. This study develops and critically evaluates a machine learning framework that predicts environmental health-related *search behavior*, used as a low-cost digital proxy rather than a measure of clinical burden, in Morocco using exclusively open-access global datasets. We integrated five years of weekly data (April 2020 to April 2025) from ERA5 meteorological reanalysis, Copernicus Atmosphere Monitoring Service products, and Google Trends to model binary high-interest states for four search categories: air quality, respiratory conditions, heat-related illness, and allergies. Ensemble models with 71 engineered features achieved strong classification performance (ROC-AUC 0.92 to 0.96) with temporal stability over 52-week horizons. An ablation study showed that this discrimination is carried primarily by the autocorrelated search signal; the environmental reanalysis block added negligible incremental ROC-AUC at national scale, yet retained meaningful standalone signal for the respiratory and air-quality categories while collapsing to near-random for allergy and heat-related illness. This contrast tracks biological plausibility, which varied substantially across categories: respiratory predictions aligned with established epidemiological patterns, whereas heat-related predictions showed weak temperature correspondence and implausible winter peaks. We therefore frame the system as a complementary screening layer pending external validation against clinical endpoints, not as a validated disease-surveillance or operational warning system. Because it relies solely on freely available data, the framework offers a replicable national-scale prototype for environmental health monitoring across North Africa and the Global South, contributing to SDG 3, 11, and 13.

1. Introduction

As global urbanization accelerates toward the 2050 targets, cities in resource-constrained settings increasingly need low-cost tools to monitor and safeguard urban public health against climate volatility, often without the ground infrastructure that established monitoring systems assume. The intersection of environmental factors and public health is therefore crucial to the modern urban agenda and the pursuit of the United Nations Sustainable Development Goals (SDGs) [1]. Rising temperatures, degraded air quality, and altered weather patterns are increasing climate-sensitive disease burdens worldwide [2, 3]. These trends threaten SDG 3 (Good Health and Well-being), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). The World Health Organization estimates that environmental risk factors contribute to approximately 24% of the global disease burden, with disproportionate impacts on low- and middle-income countries where urban monitoring infrastructure is often lacking [4].

In Morocco, these challenges are compounded by rapid urbanization, limited healthcare infrastructure, and insufficient early warning systems [5]. Air quality degradation in urban centers, driven by vehicular emissions, industrial activities, and desert dust events, has been linked to increased respiratory diseases [6, 7]. Similarly, heat-related illnesses have increased in frequency and severity, particularly affecting vulnerable populations [8]. These compound risks emphasize the urgent need for innovative, data-driven urban analytics to monitor and mitigate environmental health impacts.

To tackle these limitations, researchers are moving beyond traditional surveillance methods, which often suffer from delayed reporting and incomplete data coverage in resource-constrained settings [9]. Machine learning offers a scalable alternative for urban intelligence systems. Methods including Random Forest [10], Gradient Boosting [11], and XGBoost [12] have demonstrated superior performance in capturing complex nonlinear relationships in health applications [13, 14, 15].

Concurrently, digital epidemiology has emerged as a complement to traditional systems by leveraging internet-based behavior as a proxy for health outcomes [16]. While

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promising, this approach faces fundamental criticism regarding its susceptibility to media-driven behavior, temporal instability, and unclear correlations with actual disease burdens [17, 18, 19, 20].

To overcome these data reliability issues, high-resolution environmental data from satellite observations and reanalysis products have expanded research possibilities. The ERA5 reanalysis offers meteorological data at unprecedented resolution [21], while the Copernicus Atmosphere Monitoring Service (CAMS) provides comprehensive atmospheric composition data [22]. Validation studies have established the reliability of these satellite-derived inputs [23, 24], particularly in regions where ground-based sensing networks are sparse [25].

Although previous studies have combined subsets of these data sources, for instance, linking Google Trends with meteorology [26] or air pollution with respiratory outcomes [27, 28], important methodological gaps remain. First, reliance on random cross-validation rather than rigorous time-series splitting frequently inflates performance estimates [29, 30]. Second, the alignment between high technical performance and biological plausibility is rarely systematically assessed [17]. Third, model stability under operational forecasting conditions and proper probability calibration for risk-based decision-making remain inadequately addressed [31, 32].

This study addresses these interconnected gaps by developing a robust machine learning framework for predicting environmental health-related search patterns in Morocco. By integrating granular meteorological and atmospheric data with digital health signals, we create predictive models for air quality-related conditions, respiratory illnesses, heat-related illnesses, and allergies. To disentangle the respective predictive contributions of environmental exposures and digital search behavior, we pair the modeling with a systematic ablation analysis and a tiered biological plausibility assessment. This dual lens allows us to distinguish statistically accurate predictions from those with genuine epidemiological coherence, and to clarify what the models are actually learning. The primary contributions of this research are:

- **Rigorous Temporal Validation:** We implement an expanding-window strategy across 26, 39, and 52-week horizons to ensure the system's reliability for operational deployment [30, 31];
- **Advanced Feature Engineering:** We identify key predictive drivers by incorporating temporal encodings [33], lag structures [34], and seasonal anomaly detection [35];
- **Reliability & Calibration:** We ensure model outputs are actionable for decision-makers by implementing proper probability calibration for risk assessment [32]; and
- **Biological Plausibility:** We critically evaluate predictions against established environmental health relationships to determine the validity of digital signals as proxies for actual health burdens in urban environments [17].
- **Systematic Source Attribution:** We isolate the relative contribution of environmental reanalysis and digital search signals through an ablation study, clarifying the primary drivers of predictive performance.

By addressing both technical feasibility and validity, this work offers a replicable methodology for data-driven urban health monitoring in low- and middle-income countries. It establishes benchmarks for distinguishing accurate predictions from epidemiologically meaningful forecasts and demonstrates how open-access data can support urban environmental health monitoring aligned with SDGs 3, 11, and 13 across the Global South.

The remainder of this paper is organized as follows: Section 2 presents the materials and methods, Section 3 describes the experimental setup and results, Section 4 provides a rigorous discussion of findings, and Section 5 concludes with key takeaways and future directions.

2. Materials and Methods

This study develops a machine learning framework for predicting environmental health search patterns in Morocco using freely available global datasets. Following established best practices for environmental health prediction [14, 28], the framework consists of: (1) data acquisition from ERA5 weather reanalysis, CAMS air quality products, and Google Trends; (2) feature engineering incorporating temporal patterns, environmental exposures, interaction terms, and lag features; (3) model training with time series cross-validation, probability calibration, and threshold optimization; and (4) comprehensive evaluation using multiple metrics and expanding-window backtesting. Figure 1 provides a comprehensive overview of this pipeline.

2.1. Study Area

Located in northwestern Africa (21°/36°N, -1°/-17°W), Morocco covers approximately 710,850 km² with a population exceeding 36 million, of which 64% resides in major urban centers [36]. The country exhibits diverse climatic conditions ranging from Mediterranean in the north to arid desert in the south, with the Atlas Mountains creating distinct environmental zones [37]. Morocco faces significant environmental health challenges including air pollution, periodic Saharan dust intrusions, and increasing extreme heat events [38, 39], making it an ideal setting for developing predictive environmental health models in a resource-constrained context.

2.2. Data Collection and Integration

This stage encompasses data acquisition from multiple sources [28], temporal harmonization, and construction of

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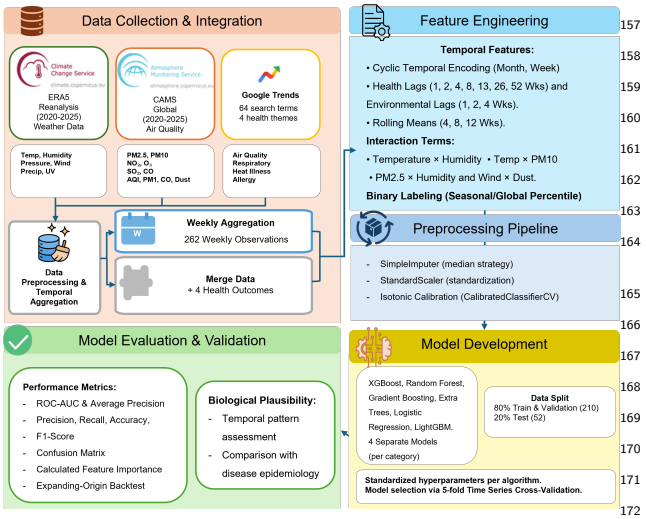


Figure 1: Methodology pipeline showing data integration, feature engineering, model development, and evaluation framework. The pipeline proceeds left-to-right: multi-source data collection (ERA5, CAMS, Google Trends) feeds into a comprehensive feature engineering stage producing 68-71 features per category, followed by model development using six algorithms with time series cross-validation, and finally a dual evaluation framework assessing both technical performance and biological plausibility.

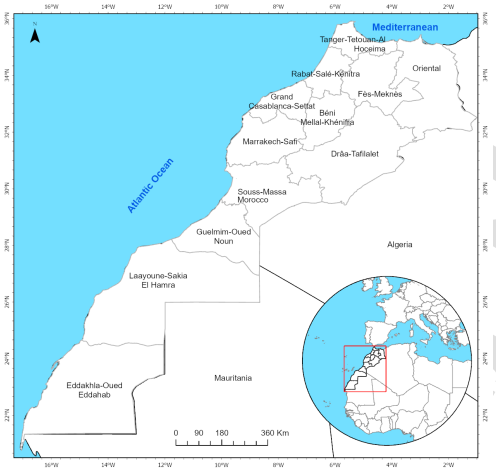


Figure 2: Study area map showing Morocco's geographical location, and administrative regions included in the environmental health prediction analysis.

unified weekly time series [40] spanning five years (April 2020-April 2025, 262 weeks) at national level for Morocco.

2.2.1. Data Sources

We integrated three primary data sources providing complementary information on environmental exposures and health outcomes:

- **ERA5 Climate Reanalysis:** Meteorological data from ECMWF ERA5 reanalysis [21] provided hourly atmospheric variables at approximately 31 km spatial resolution, including temperature (2-meter air temperature), dewpoint temperature, surface pressure, mean sea level pressure, 10-meter wind components (u and v), total cloud cover, total precipitation, and surface solar radiation. Additional variables were derived including relative humidity (Magnus formula), wind speed, UV index (from solar radiation), and heat index.

- **CAMS Air Quality Data:** Air quality data from Copernicus Atmosphere Monitoring Service (CAMS) global reanalysis [22] provided hourly concentrations of particulate matter (PM1, PM2.5, PM10), including PM1 due to its enhanced capability to penetrate deep into the alveolar region and enter the bloodstream relative to coarser particles [41], trace gases (CO, NO_2, O_3, SO_2), dust aerosol optical depth, and total aerosol optical depth, with US EPA Air Quality Index calculated from PM2.5 [42]. We note that although PM1 was included for its physiological relevance, it did not appear among the ranked predictive features for any category (Section 3), indicating a negligible marginal contribution beyond PM2.5 and PM10 at this aggregation scale; we retain it for completeness and transparency.

- **Google Trends Health Signals:** Health-related search interest data were collected from Google Trends for Morocco, covering 64 search terms across English, French, and Arabic organized into four health categories: respiratory illnesses (28 terms), allergies (18 terms), air quality concerns (9 terms), and heat-related illnesses (14 terms). Google Trends provides weekly search interest normalized to 0-100 scale [26], with category-level indicators computed as mean search interest across constituent terms.

These three data sources, operating at different temporal resolutions and spatial scales, required systematic harmonization before integration.

2.2.2. Temporal Aggregation and Alignment

Given the different temporal resolutions and reference frames across data sources, we implemented a two-step harmonization procedure following standard practices in environmental epidemiology [43].

In the first stage, hourly data were aggregated to daily values using means for temperature, humidity, pressure, wind speed, cloud cover, and UV index, and sums for precipitation. In the second stage, daily values were aggregated to weekly means to match the Google Trends temporal resolution. Temperature additionally retained weekly minimum and maximum values to capture within-week variability.

A critical alignment step addressed differing week definitions: Google Trends reports weeks ending Saturday, while initial weather/air quality aggregations used Monday week starts (ISO 8601 standard). We converted all environmental data to Saturday-ending weeks by adding five days to date

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stamps, ensuring environmental exposures and health search patterns correspond to identical weekly periods. This alignment is essential for establishing valid exposure-outcome relationships. The complete aggregation and alignment workflow is illustrated in Figure 3.

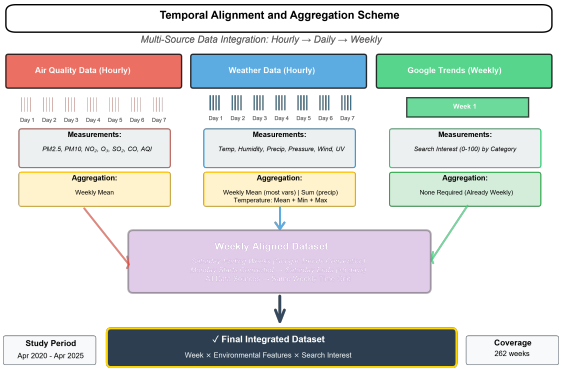


Figure 3: Temporal aggregation and alignment scheme showing multi-source data integration. Hourly air quality and weather data are first aggregated to daily values, then to weekly means matching Google Trends' Saturday-ending weekly resolution. The alignment step ensures that environmental exposures and health search patterns correspond to identical weekly periods, which is essential for valid exposure-outcome modeling.

Beyond temporal alignment, we aggregated gridded environmental data to a national scale to match the spatial granularity of the health outcome data. While averaging meteorological and pollutant variables across the country's diverse geography may attenuate local extremes (e.g., urban heat islands or localized industrial plumes), this approach was necessitated by the resolution of the Google Trends health signals, which are most robust and representative at the national level for this region [20, 26]. This trade-off prioritizes the stability and volume of the digital health signal over local environmental precision, consistent with established methods in digital disease surveillance [16].

This decision was further supported by empirical assessment of sub-national Google Trends data availability. We queried the same allergy-related search terms used in our study at the regional level for Grand Casablanca, the country's most populous region, and found substantial data sparsity even at monthly resolution: of the 8 allergy terms tested over the study period, only 2 high-volume terms (one French, one Arabic) retained full monthly coverage, while the remaining 6 terms showed data for fewer than 48% of months, with several dropping below 3%. If data sparsity at monthly resolution is this severe in the largest urban region, weekly modeling at sub-national scales, as required by our framework, would be infeasible for the majority of search terms. National aggregation mitigates this limitation by pooling search volumes across regions, producing more stable and representative time series at the cost of spatial granularity. Future spatial disaggregation efforts should therefore prioritize integration with alternative data sources (e.g., social media, telecom mobility data) rather than relying solely on sub-national Google Trends.

Consequently, the system presented here is a national-scale (and, in principle, regional-scale) prototype rather than a city-scale operational platform. City-level modeling would require sub-national data sources that, as shown above, Google Trends cannot currently supply at weekly resolution for Morocco.

The aligned data sources were then merged to create an integrated dataset for analysis.

2.2.3. *Data Merging and Verification*

Following this harmonization procedure, we merged all data sources on the standardized Saturday-ending weekly dates, creating an integrated dataset with 262 weekly observations. Data completeness was verified with no missing values across all sources for the study period, reflecting the advantage of satellite-derived reanalysis products in regions with sparse ground-based monitoring infrastructure [21, 22]. The final merged dataset contained weather variables, air quality indicators, and four health outcome time series ready for feature engineering and model development.

2.2.4. *Target Variable Construction*

We constructed binary classification targets for the four health categories using two labeling strategies based on temporal characteristics.

For seasonal categories (allergies, heat-related illnesses), we applied within-month percentile thresholds: for each calendar month separately, the 70th percentile of search interest across all years was computed. Weeks exceeding this month-specific threshold were labeled as **high search interest periods** (seasonal class = 1) and weeks below as **low search interest periods** (seasonal class = 0) [40]. This specific threshold was selected as a methodological trade-off: it is sufficiently rigorous to isolate significant "outbreak" signals from background noise, yet broad enough to retain adequate positive samples for model training, avoiding the severe class imbalance associated with higher statistical cutoffs (e.g., 90th percentile) [18].

To validate this threshold choice, we conducted a sensitivity analysis evaluating the 60th, 70th, 80th, and 90th percentiles for seasonal categories (Table 1). At the 60th percentile, both categories retain approximately 41% positive samples, but the broader positive class captures substantial background noise, reducing the clinical relevance of the "high interest" label for early warning applications. Although the 60th percentile yields nominally higher F1-scores (allergy: 0.784; heat-related: 0.819), this reflects the model's ability to predict a less stringent threshold rather than its ability to detect meaningful health signal elevations. At the 80th percentile, positive sample proportions drop to 22.5% and 23.7% for allergies and heat-related illness respectively, with F1-scores of 0.696 and 0.805. At the 90th percentile, where only 13.4% of weeks are positive, performance declines further (allergy F1 = 0.536, heat-related F1 = 0.730). The 70th percentile thus represents the optimal

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Table 1
Sensitivity analysis of percentile thresholds for seasonal categories. Cross-validation F1-scores and positive class proportions are shown for each threshold. The 70th percentile (bold) was selected as the optimal trade-off between signal specificity and training stability.

Threshold	Allergy		Heat-Related	
	% Positive	CV F1	% Positive	CV F1
60th percentile	40.8	0.784	41.2	0.819
70th percentile	**32.1**	**0.713**	**32.4**	**0.789**
80th percentile	22.5	0.696	23.7	0.805
90th percentile	13.4	0.536	13.4	0.730

trade-off: it isolates genuinely elevated search periods while maintaining sufficient positive samples ($\approx$ 32%) for stable model training, avoiding the severe class imbalance that degrades performance at higher thresholds.

For non-seasonal categories (air quality, respiratory), we employed a global median split: the median search interest across the entire study period was computed, with weeks above this threshold labeled as **high search interest periods** (non-seasonal class = 1) and weeks at or below labeled as **low search interest periods** (non-seasonal class = 0).

We employed a binary classification framework rather than regression to enhance robustness against normalization artifacts. Since the search index is relative rather than absolute, precise values can fluctuate due to unrelated search behavior; mapping these to binary states focuses the model on detecting meaningful signal elevations, the primary objective of public health early warning systems [17, 9].

These labeling strategies resulted in balanced to moderately imbalanced class distributions across the full 262-week dataset: air quality (51.5% high search interest, 135 high / 127 low), respiratory (50.0% high search interest, 131 high / 131 low), allergies (32.1% high search interest, 84 high / 178 low), and heat-related illness (32.4% high search interest, 85 high / 177 low). These proportions refer to the complete study period; the held-out test set (final 53 weeks) has a somewhat different composition, with positive-class shares of 54.7% (air quality), 54.7% (respiratory), 26.4% (allergies), and 39.6% (heat-related illness). The binary labels enable supervised machine learning while preserving temporal ordering for proper time series validation.

2.3. Feature Engineering and Preprocessing

We developed a comprehensive 68-71 feature set (varying by category due to category-specific health search patterns) integrating temporal patterns, environmental variables, interaction terms, and historical health signals. All feature engineering respected temporal ordering to prevent information leakage [29].

2.3.1. Temporal Features

We created cyclical encodings using sine and cosine transformations to capture periodic patterns in time series data [44, 33]. For monthly patterns, we encoded the month of year m (ranging from 1 for January to 12 for December) as follows:

$$\begin{cases} \text{month_sin} = \sin\left(\frac{2\pi m}{12}\right) \\ \text{month_cos} = \cos\left(\frac{2\pi m}{12}\right) \end{cases}$$

where m represents the calendar month (1-12), 12 is the total number of months in a year, month_sin captures the sine component of the monthly cycle (peaking in spring, lowest in fall), and month_cos captures the cosine component (peaking in winter, lowest in summer). Together, these two features uniquely identify each month's position in the annual cycle.

For weekly patterns, we encoded the week of year w (ranging from 1 to 52) as follows:

$$\begin{cases} \text{week_sin} = \sin\left(\frac{2\pi w}{52}\right) \\ \text{week_cos} = \cos\left(\frac{2\pi w}{52}\right) \end{cases}$$

where w represents the week of year (1-52), 52 is the total number of weeks in a year, week_sin captures the sine component of the weekly annual cycle, and week_cos captures the cosine component. The combination of sine and cosine allows the model to learn any phase of seasonal variation.

These cyclical encodings ensure temporal continuity across year boundaries, avoiding discontinuities between December and January or between week 52 and week 1. We also included one-hot encoded season indicators (Spring, Summer, Fall, Winter), yielding 7 temporal features total.

In addition to temporal patterns, environmental variables were incorporated as predictive features.

2.3.2. Environmental Features

While temporal features capture seasonal patterns, environmental variables provide the mechanistic link between atmospheric conditions and health outcomes. We incorporated 23 current-week environmental variables organized into two categories, as summarized in Table 2. These multi-scale

Table 2
Environmental features incorporated in the prediction models.

Category	Variables
Weather (12)	Temperature (mean, max, min), humidity, wind speed, cloud cover, precipitation, pressure, UV index, heat index, extreme indicators
Air Quality (11)	PM1, PM2.5, PM10, NO_2, O_3, CO, SO_2, dust AOD, total AOD, AQI
Lags (21)	1, 2, 4-week lags for: temperature, humidity, PM2.5, PM10, AQI, NO_2, O_3
Total	**44 features**

lags reflect both immediate physiological responses (1-week lag) and cumulative inflammatory effects from prolonged

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exposure (2-4 week lags) [45]. The following subsection describes interaction terms engineered to capture combined atmospheric effects.

2.3.3. Interaction Features

Beyond individual environmental effects, certain atmospheric conditions interact synergistically to amplify health impacts. We engineered four interaction terms capturing these combined effects [46, 47], as summarized in Table 3.

Table 3
Interaction features capturing synergistic environmental effects.

Formula	Health Consequence	Reference
T × H / 100	Combined heat-humidity stress reduces thermoregulation capacity	[48, 49]
$PM_{2.5}$ × H / 100	Hygroscopic particle growth increases respiratory penetration	[50]
T × PM_{10} / 100	Heat enhances dust resuspension and inhalation risk	[51]
W × AOD_d / 10	Wind-driven dust mobilization increases airborne particulates	[6]

where T = mean temperature (°C), H = relative humidity (%), $PM_{2.5}$ = fine particulate matter concentration ($\mu g/m^3$), PM_{10} = coarse particulate matter concentration ($\mu g/m^3$), W = wind speed (km/h), and AOD_{dust} = dust aerosol optical depth. Scaling factors (100 and 10) normalize interaction magnitudes to be comparable with other features.

These interaction terms capture non-additive environmental effects that single variables cannot represent. The following subsection describes how historical health search patterns were incorporated to capture temporal dependencies.

2.3.4. Health Search Features

While environmental lags capture delayed physiological responses to atmospheric exposures, health search patterns exhibit their own temporal dependencies through autocorrelation and seasonality. To capture these behavioral patterns, we engineered features from historical search data as summarized in Table 4 [52, 33].

Table 4
Health search features capturing temporal dependencies.

Feature Type	Description	Count
Value lags	Search interest at 1, 2, 4, 8, 13, 26, 52 weeks prior	7
Class lags	Binary high/low labels at same time points	7
Rolling means	Moving averages over 4, 8, 12-week windows	3
Seasonal anomaly	Deviation from week-of-year baseline	1
Total		**18**

The 52-week lag specifically captures year-over-year seasonal recurrence, while rolling means (calculated excluding current week to prevent leakage) capture medium-term trends. For seasonal anomaly detection, we computed:

$$\left\{ \text{anomaly}_t = S_t - \widetilde{S}_{w(t)} \right.$$

where S_t denotes the observed search interest at time t, $\widetilde{S}_{w(t)}$ is the median search interest for the corresponding week-of-year $w(t)$ computed from training data only, and $w(t)$ maps time t to its week-of-year (1-52). Positive anomalies indicate higher-than-expected search interest relative to seasonal norms, while negative values indicate lower-than-expected interest. Computing medians exclusively from training data prevents information leakage [35].

These health search features, combined with temporal and environmental variables, form the complete feature set summarized in the following subsection.

2.3.5. Feature Set Summary

The preceding subsections detailed the engineering of temporal, environmental, and health search features. Table 5 summarizes the complete feature architecture, totaling 68-71 features per category depending on category-specific health search patterns.

Table 5
Feature engineering summary by category with exact counts.

Feature Category	Description	Count
Temporal Encodings	Cyclical month/week transformations (i.e., `month_sin`, `month_cos`, `week_sin`, `week_cos`) and season indicators	7
Environmental (Current)	Weather variables (i.e., `avgtempC`, `humidity`, `windspeedKmph`) + Air Quality variables (i.e., `PM2_5`, `NO2`, `AQI`)	23
Environmental Lags	Lagged exposures at 1, 2, 4 weeks (i.e., `avgtempC_lag1`, `PM2_5_lag4`) for 7 key variables	21
Environmental Interactions	Synergistic effects (i.e., `temp_humidity`, `pm25_humidity`)	4
Health Search Value Lags	Historical search interest at 1, 2, 4, 8, 13, 26, 52 weeks (i.e., `respiratory_lag1`, `allergy_lag52`)	7
Health Search Class Lags	Historical binary classifications (i.e., `respiratory_class_lag1`)	7
Rolling Window Features	Moving averages over 4, 8, 12 weeks (i.e., `roll4_mean`, `roll12_mean`)	3
Seasonal Anomaly	Deviation from training baseline (i.e., `seasonal_anomaly`)	1
Total (per category)	**Category-specific total**	**68-71**

Prior to model training, all features underwent standardized preprocessing as described in the following subsection.

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2.3.6. Preprocessing Pipeline

All variables required standardized preprocessing before model training, implemented using scikit-learn 1.5.1 [53]. Initial column-wise preprocessing replaced infinite values with missing indicators, followed by forward filling and median imputation for continuous features, and forward filling followed by zero-filling for binary class lags. Features were then processed through a three-step pipeline: (1) median imputation for remaining missing values including structural missingness from lag features in early periods; (2) standardization to zero mean and unit variance using $z = (x - \mu)/\sigma$; and (3) the classification model. Imputer and scaler parameters were fit exclusively on training data and applied to test data to prevent information leakage [54]. This pipeline was wrapped inside a probability calibration wrapper (`CalibratedClassifierCV`) to ensure calibration was performed on preprocessed features.

The following subsection describes the model development process, including sample weighting strategies for seasonal categories.

2.4. Model Development

This study employs a supervised machine learning approach for binary classification of health search interest periods. The modeling strategy comprises three core components: (i) sample weighting to address seasonal imbalances in health categories with strong temporal patterns, (ii) probability calibration to ensure reliable risk estimates beyond simple class predictions, and (iii) threshold optimization to balance precision and recall according to each category's characteristics.

2.4.1. Sample Weighting for Seasonal Categories

For categories with strong seasonal patterns (allergies, heat-related illnesses), we applied sample weights during model training to emphasize in-season observations [40, 55]. Allergy observations during spring and fall seasons (March-May, August-October) received weight 1.5 versus 1.0 for other months. Heat-related illness observations during summer (June-September) received weight 1.5 versus 1.0 for other months.

The specific weighting factor of 1.5 was selected through an empirical grid search on the training data. We evaluated weights ranging from 1.0 to 2.0 and found that a factor of 1.5 optimized the trade-off between sensitivity (detecting true peaks) and precision (avoiding false alarms during off-peak transitions), whereas higher weights led to overfitting on seasonal noise [56].

Air quality and respiratory categories used uniform weights (1.0). Sample weights were applied during all training procedures using scikit-learn's sample weighting parameter (`sample_weight`) [53].

The implementation of probability calibration, threshold optimization, and complete experimental procedures, including train-test splitting, model selection, and evaluation framework, is detailed in Section 3.

3. Experimental Setup and Results

All experiments were conducted on a Lenovo workstation equipped with an Intel Core i7-9750H processor (2.60 GHz, 12 CPUs), 16 GB RAM, and NVIDIA GeForce GTX 1650 GPU (4 GB VRAM), running Windows 11 Home 64-bit. The modeling pipeline was implemented in Python 3.12, leveraging scikit-learn 1.5.1 [53] for preprocessing, model training, calibration, and evaluation; XGBoost 3.2.0 [12] and LightGBM 4.6.0 [57] for gradient boosting implementations; pandas 2.2.2 and NumPy 1.26.4 for data manipulation; and Matplotlib 3.9.2 for visualization. All random processes used seed 42, and the complete pinned environment together with a one-command reproduction script is provided in the public repository (see Data and Code Availability), so that all reported results, tables, and figures can be regenerated exactly.

3.1. Train-Test Split and Cross-Validation

Following best practices for time series validation [29, 30], we employed temporal train-test split, reserving the final 20% of weeks (52 weeks) as held-out test set, with remaining 80% (210 weeks) for training. For model selection, we used time series cross-validation with 5 folds (`TimeSeriesSplit`) creating chronologically ordered train-validation splits where each successive fold uses an expanding training set and the next chronological segment for validation [33], preventing information leakage from future to past (Figure 4).

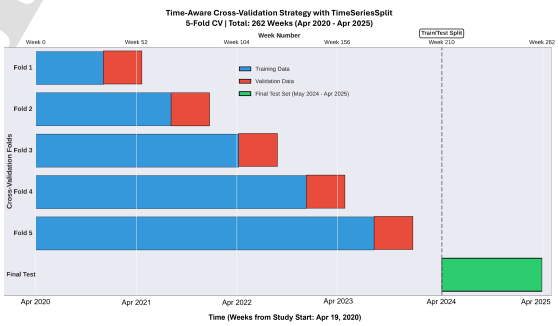


Figure 4: Time series cross-validation strategy showing the expanding window approach across 262 weeks. Each fold uses all previous data for training (blue) and the next chronological segment for validation (red), ensuring that no future information leaks into training. The final test set (green, last 52 weeks from April 2024 to April 2025) is held out entirely from model selection and used only for final evaluation.

3.2. Model Algorithms and Fixed Hyperparameters

We evaluated six machine learning algorithms representing different learning paradigms [58, 59], implemented using scikit-learn 1.5.1, XGBoost 3.2.0, and LightGBM 4.6.0 with fixed random seeds (42). Rather than performing exhaustive hyperparameter search, we utilized standardized,

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fixed configurations for each algorithm to ensure fair comparison of their inherent strengths [59]: (1) Logistic Regression with L2 regularization (C=1.0) [60], (2) Random Forest (400 trees, unlimited depth, min 5 samples/split) [10], (3) Extra Trees (600 trees, min 4 samples/split) [61], (4) Gradient Boosting (300 trees, learning rate 0.05, max depth 3) [11], (5) XGBoost (400 trees, max depth 4, learning rate 0.05, L2 regularization) [12], and (6) LightGBM (500 trees, leaf-wise growth, learning rate 0.05, max 63 leaves) [57].

3.3. Probability Calibration and Threshold Optimization

To ensure reliable probability estimates essential for risk assessment, all models underwent isotonic regression calibration using `CalibratedClassifierCV` with 3-fold cross-validation on training data [62, 32]. The preprocessing pipeline (imputation + scaling + model) was wrapped inside the calibrator, ensuring calibration probabilities were learned on preprocessed features. Isotonic regression provides non-parametric monotonic mapping from uncalibrated scores to calibrated probabilities, particularly effective for tree-based methods producing poorly calibrated probabilities [32].

Rather than using default 0.5 probability threshold, we optimized thresholds to maximize F1-score on cross-validation data [63, 64]. For each candidate threshold from the precision-recall curve (clipped to [0.02, 0.98]), we computed predicted positive rate and filtered to thresholds producing rates between 5-95%. Among valid thresholds, we selected that maximizing F1-score, falling back to Youden's J statistic ($sensitivity + specificity - 1$) from ROC curve if no thresholds achieved F1 > 0. This data-driven selection balances precision and recall according to each category's validation characteristics [65].

3.4. Cross-Validation Model Selection

For each health category and algorithm, we: (1) performed 5-fold time series cross-validation, fitting calibrated models on fold training data with sample weights and generating predictions on fold validation data, (2) concatenated all validation predictions and optimized classification threshold on combined data using F1-maximization, and (3) computed cross-validation metrics (ROC-AUC, Average Precision, Accuracy, F1-score). Models were ranked by cross-validation ROC-AUC (primary) and F1-score (secondary) [66, 59]. The best-performing model for each category was selected for final training on complete training set and evaluation on held-out test set.

Following model selection, the best-performing algorithm for each category was retrained on the complete training set using the full preprocessing and calibration pipeline described in Section 2, with optimized thresholds applied to generate final test set predictions.

3.5. Model Evaluation Framework

Evaluating classification model performance requires multiple complementary metrics that capture different aspects of predictive quality [67, 68]. This subsection presents the performance metrics used to assess model discrimination, calibration, and classification accuracy.

3.5.1. Performance Metrics

Table 6 presents the confusion matrix structure used in this study.

Table 6
Confusion matrix and derived performance measures.

	Observed Classes		
Predicted Classes	Positive	Negative	Measures
Positive	True Positive (TP)	False Positive (FP)	Precision (PPV)
Negative	False Negative (FN)	True Negative (TN)	NPV
Measures	Sensitivity (Recall)	Specificity	Accuracy

We computed standard threshold-dependent metrics including Precision, Recall, Specificity, Negative Predictive Value, Accuracy, and F1-Score as the harmonic mean of Precision and Recall [67, 68, 69, 70]. Beyond these, we employed threshold-independent measures: **ROC-AUC** quantifies overall discrimination ability across all classification thresholds [71, 72], while **Average Precision** summarizes the precision-recall curve, particularly informative for imbalanced datasets [73]. Additionally, **calibration curves** assessed agreement between predicted probabilities and observed frequencies [32, 74].

To quantify uncertainty in performance estimates, we computed bootstrap confidence intervals for all test-set metrics. For each metric, we generated 1,000 bootstrap resamples of the test set predictions (with replacement), computed the metric on each resample, and reported the 2.5th and 97.5th percentiles as the 95% confidence interval. This non-parametric approach accounts for sampling variability in the held-out test set without distributional assumptions. The same procedure was applied to expanding-window backtesting results to assess the robustness of temporal stability findings.

3.5.2. Feature Importance Methodology

To understand feature contributions to predictions [75], we report *permutation importance* [10], computed on the held-out test set (10 repeats, ROC-AUC scoring) for the selected model of each category. Permutation importance was chosen over model-internal (Gini/gain) scores because it measures the actual loss in test-set discrimination when a feature is randomly shuffled, and is therefore comparable across the different model families used here (Random Forest for air quality, Gradient Boosting for allergy and respiratory, XGBoost for heat-related illness) and less biased toward high-cardinality features than impurity-based measures. Importances were computed from non-calibrated pipelines fitted on the training data and are reported as the

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mean decrease in ROC-AUC across the 10 permutations; Figure 9 displays the ten highest-ranked features per category.

3.5.3. *Temporal Validation: Expanding-Window Backtesting*

To assess model robustness and temporal stability beyond single train-test split, we performed expanding-window backtesting on three horizons: 26, 39, and 52 weeks [30, 31]. For each horizon h, we defined cutoff point (total data length minus h weeks), split data (training = all data before cutoff; testing = last h weeks), trained calibrated model on pre-cutoff data with sample weights, optimized threshold on last 20% of training data, generated predictions on post-cutoff test period, and computed performance metrics (ROC-AUC, Average Precision, Accuracy, Precision, Recall, F1-score). This expanding-window validation provides realistic assessment of operational performance, revealing potential issues with temporal drift [30].

3.6. Cross-Validation Performance Comparison

Table 7 summarizes cross-validation performance across all six algorithms evaluated. Tree-based ensemble methods showed the highest scores, with Gradient Boosting and XGBoost showing the highest average performance. Logistic Regression showed lower performance with average AUC of 0.731.

Table 7
Cross-Validation Performance by Algorithm (Average across categories)

Algorithm	Mean ROC-AUC	Mean Accuracy
Gradient Boosting	0.928	0.832
XGBoost	0.925	0.829
Random Forest	0.919	0.821
Extra Trees	0.908	0.805
LightGBM	0.901	0.798
Logistic Regression	0.731	0.652

The performance gap between ensemble methods (mean AUC: 0.90-0.93) and Logistic Regression (AUC: 0.73) was consistent across all four health categories.

3.7. Overall Performance Summary

Table 8 presents the comprehensive performance metrics for all four health category models on the held-out test set, with 95% bootstrap confidence intervals. Our binary classification models achieved ROC-AUC scores ranging from 0.924 to 0.957 and F1-scores from 0.837 to 0.923 across all categories.

Gradient Boosting achieved the best performance for allergy and respiratory categories, while Random Forest and XGBoost performed best for air quality and heat-related illness, respectively.

Figure 5 illustrates the error distributions underlying these metrics. Among all models, the respiratory model achieved the highest recall (0.966), correctly identifying 28 of 29 high-interest periods with a single false negative

Table 8
Overall Performance Metrics by Model Type with 95% Bootstrap Confidence Intervals

Metric	AQ (RF)	Allergy (GB)	Heat (XGB)	Respiratory (GB)
ROC-AUC	0.947 [0.88-0.99]	0.957 [0.88-1.00]	0.924 [0.83-0.99]	0.936 [0.86-0.99]
Precision	0.833	1.000	0.818	0.800
Recall	0.862	0.857	0.857	0.966
F1-Score	0.847 [0.73-0.94]	0.923 [0.80-1.00]	0.837 [0.69-0.94]	0.875 [0.78-0.95]
Accuracy	0.830	0.962	0.868	0.849

(the week of 27 April 2024), whose predicted probability fell just below the decision threshold. We verified that this high sensitivity was not an artifact of class imbalance, as the test set was approximately balanced (54.7% positive, 29 of 53 weeks), and thresholds were optimized to maximize the F1-score rather than recall alone; the ablation analysis in Section 3.12 further confirms that comparable recall is attainable without search-history features only by collapsing discrimination. In contrast, the allergy model prioritized precision (1.000) over recall, producing zero false positives among 39 negative cases while maintaining recall at 0.857 (2 false negatives among 14 positives). The heat-related illness model achieved 0.857 recall with 4 false positives and 3 false negatives, while the air quality model exhibited a balanced error distribution, producing 25 true positives, 4 false negatives, 5 false positives, and 19 true negatives.

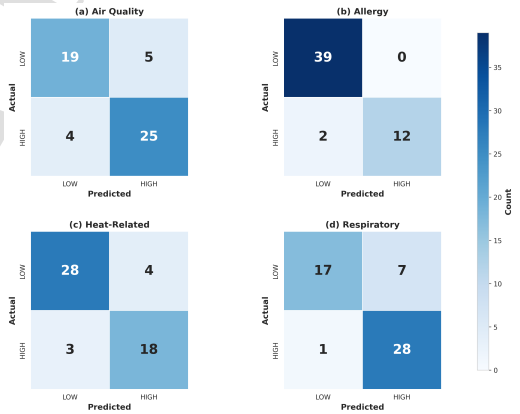


Figure 5: Confusion matrices showing classification performance on the held-out test set (53 weeks). Each cell displays the count of observations classified into each category. True positives (upper-right) and true negatives (lower-left) represent correct predictions, while false positives and false negatives indicate classification errors. The respiratory model achieved the highest recall (28 of 29 high-interest weeks correctly identified, one false negative).

3.8. Discrimination and Calibration Performance

Figure 6 displays the ROC curves for all four health category models. The allergy model achieved the highest AUC (0.957), followed by the air quality model (0.947) and the respiratory model (0.936), with heat-related illness at 0.924. All curves show strong separation from the diagonal reference line.

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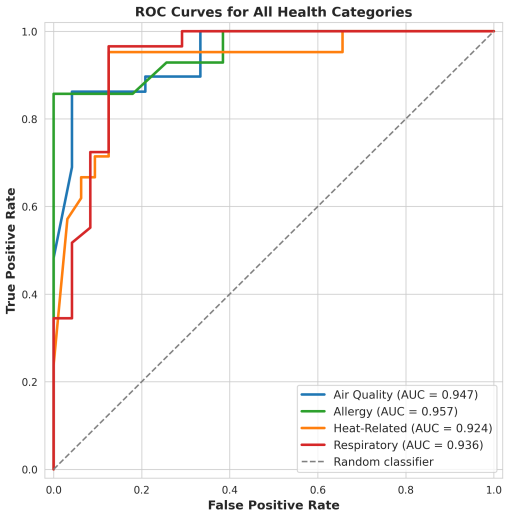


Figure 6: ROC curves for all four health category models, plotting true positive rate (sensitivity) against false positive rate (1 − specificity) across all classification thresholds. The diagonal dashed line represents a random classifier (AUC = 0.5); greater area between the model curve and the diagonal indicates stronger discriminative ability. All models achieve AUC values above 0.94, indicating excellent separation between high-risk and low-risk weeks.

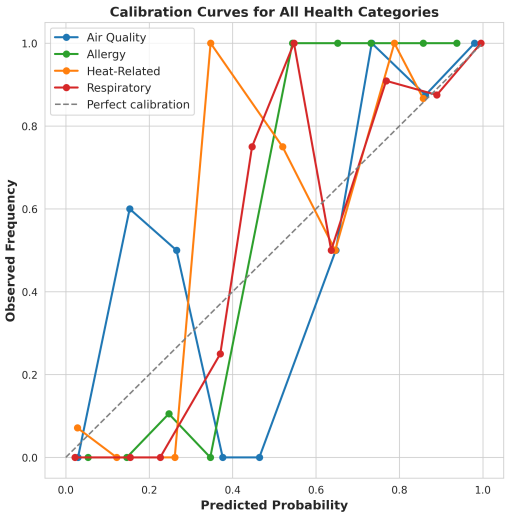


Figure 7: Calibration curves after isotonic calibration for all four health category models. The diagonal line represents perfect calibration, where predicted probabilities exactly match observed frequencies. Points above the diagonal indicate under-confidence (predicted probability lower than actual frequency), while points below indicate over-confidence. Allergy and respiratory models track the diagonal most closely, suggesting well-calibrated probability outputs suitable for risk-based decision-making. Calibration curves were computed using 10 uniformly spaced probability bins; given the modest test-set size (53 weeks), per-bin counts are small, which accounts for the visual instability in sparsely populated bins.

Calibration curves after isotonic calibration are shown in Figure 7. The allergy and respiratory models (both Gradient Boosting) track closely to the diagonal across most probability ranges. The air quality model shows deviation in low-probability regions (below 0.3) but tracks the diagonal in the 0.3 to 0.7 probability range. The heat-related model deviates slightly above the diagonal in the 0.6 to 0.7 range.

3.9. Temporal Prediction Accuracy

Figure 8 illustrates the temporal alignment between predicted probabilities and actual high-risk search periods across the 52-week test period (April 2024 to April 2025). Predicted probabilities align closely with observed high-risk periods across all categories.

The air quality model (panel A) shows two distinct clusters of elevated probabilities: late spring 2024 (April-May) and an extended period from December 2024 through April 2025. The allergy model (panel B) exhibits elevated probabilities primarily during summer months (June through September 2024), with additional isolated high-risk weeks distributed throughout the year. Heat-related predictions (panel C) show elevated probabilities during late spring 2024 (May-June) and late winter to spring 2025 (January-April), with peak probabilities during these periods. The respiratory model (panel D) shows elevated probabilities concentrated during fall and winter months (October 2024 through March 2025).

3.10. Feature Importance Results

Feature importance analysis (Figure 9) shows the relative contribution of individual features to model predictions, measured as the mean decrease in test-set ROC-AUC under permutation. Across all four categories the seasonal anomaly feature was by far the strongest single predictor, with a mean AUC-drop importance of 0.615 for allergy, 0.464 for heat-related illness, 0.228 for respiratory, and 0.060 for air quality, far exceeding any environmental or individual search-lag feature. This dominance is consistent with the ablation results (Section 3.12), in which the search-history block accounts for almost all discriminative performance.

Below the seasonal anomaly, the contributing features were category-specific but uniformly small in comparison. For respiratory, the leading sub-dominant features were the 52-week search lag (0.038), the 2-week search lag (0.034), and the 4-week search rolling mean (0.019), followed by environmental terms including the 1-week mean-temperature lag (0.016), the 4-week $PM_{2.5}$ lag (0.014), O_3, and NO_2. For air quality, the 2-week NO_2 lag (0.006), the temperature×PM_{10} interaction (0.005), and the 4-week mean-temperature lag (0.004) were the leading contributors alongside the 1-week search lag. For allergy and heat-related illness, no environmental or search feature approached the seasonal anomaly; the largest secondary terms (SO_2 and isolated search lags) contributed mean AUC drops below 0.015, indicating that prediction in these two categories is driven almost entirely by the anomaly and the autocorrelation it encodes. Search-history features (value lags, class lags, and rolling means) appeared among the secondary contributors

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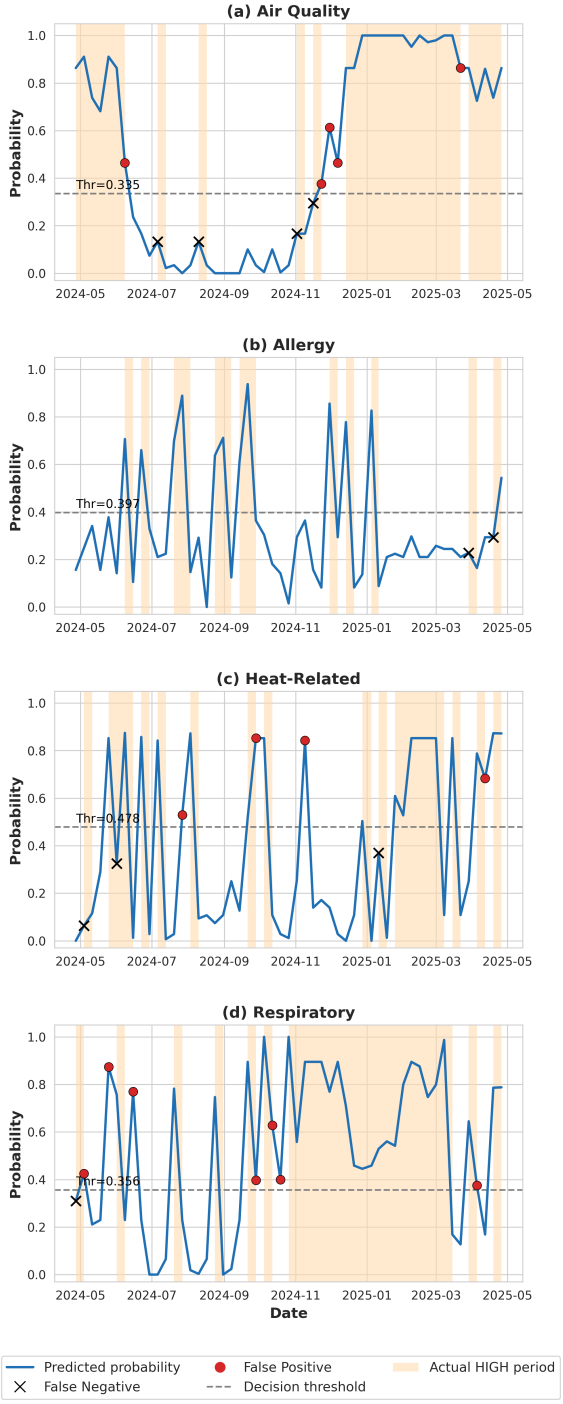


Figure 8: Temporal prediction performance across all health categories during the 52-week test period (April 2024–April 2025).

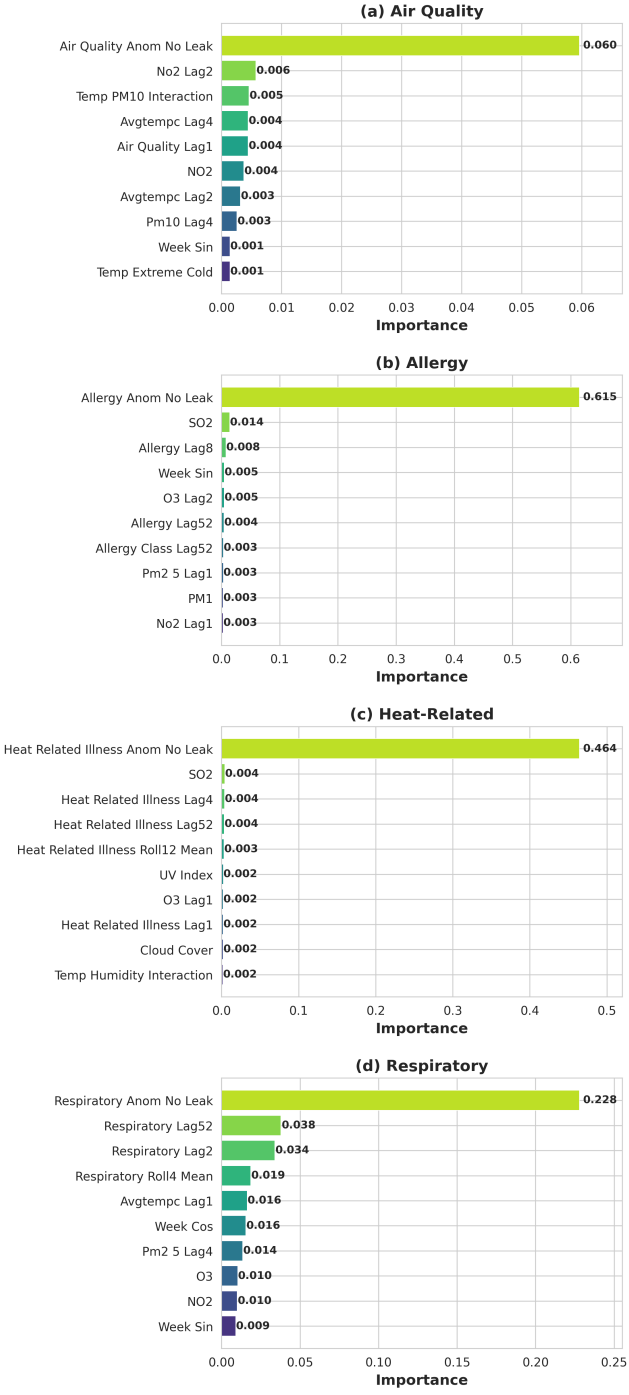


Figure 9: Top 10 predictive features for each health category, ranked by permutation importance (mean decrease in held-out test-set ROC-AUC over 10 permutations). The seasonal anomaly feature dominates every category; note the differing horizontal scales across panels.

across categories, consistent with the ablation finding that the digital signal carries the predictive load.

3.11. Expanding-Window Backtesting Results

Expanding-window backtesting evaluated model performance across three prediction horizons (Table 9). All models maintained ROC-AUC $\geq$ 0.88 across all horizons, with respiratory and heat-related models showing the highest discrimination at shorter horizons. Performance at the 52-week horizon confirmed temporal stability, with all categories achieving AUC $\geq$ 0.93 except air quality at the 26-week

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Table 9
Temporal Stability: Expanding-Window Backtesting Results (ROC-AUC with 95% CI)

Outcome	26 weeks	39 weeks	52 weeks
Air Quality	0.88 [0.76-1.00]	0.94 [0.87-1.00]	0.94 [0.88-1.00]
Allergy	0.88 [0.75-1.00]	0.93 [0.85-1.00]	0.97 [0.93-1.00]
Heat	0.95 [0.87-1.00]	0.96 [0.90-1.00]	0.93 [0.86-1.00]
Respiratory	0.99 [0.95-1.00]	0.99 [0.96-1.00]	0.94 [0.87-1.00]

horizon (AUC = 0.88), likely reflecting the limited training data available at this shortest cutoff.

Overall, this temporal stability across extended prediction windows, combined with the strong discrimination and calibration performance reported above, supports the use of web search behavior as a complementary, cost-effective signal for anticipating environmental health risks at the population level, pending external validation against clinical outcomes.

3.12. Ablation Study and Baseline Comparison

To isolate the contribution of the ERA5/CAMS environmental block from the temporal autocorrelation already present in the search series, we evaluated five feature configurations per category, holding the algorithm fixed to the published selection so that differences reflect features rather than model choice (Table 10). We additionally report two trivial baselines (Table 11) and a compact value-add decomposition (Table 12).

Table 10
Ablation across feature configurations (held-out test set, 95% bootstrap CIs). Algorithm fixed per category to the published selection. The "Temporal+Env" row removes all search-history features to probe the respiratory model's recall. The recall of 1.000 in several Env_only, Temporal+Env, and Temporal_only rows reflects degenerate near-constant positive prediction, as the accompanying low F1 and collapsed ROC-AUC confirm, rather than genuine detection.

Category	Configuration	Feat.	ROC-AUC [95% CI]	F1	Recall
Air Quality	Full	71	0.947 [0.88-0.99]	0.847	0.862
	Temporal+Search	25	0.958 [0.90-0.99]	0.862	0.862
	Env_only	46	0.912 [0.81-0.99]	0.844	0.931
	Temporal+Env	53	0.901 [0.79-0.99]	0.844	0.931
	Temporal_only	7	0.894 [0.78-0.98]	0.852	0.897
Allergy	Full	71	0.957 [0.88-1.00]	0.923	0.857
	Temporal+Search	25	0.976 [0.92-1.00]	0.839	0.929
	Env_only	46	0.432 [0.28-0.58]	0.418	1.000
	Temporal+Env	53	0.462 [0.31-0.62]	0.418	1.000
	Temporal_only	7	0.403 [0.21-0.59]	0.418	1.000
Heat-Related	Full	71	0.924 [0.83-0.99]	0.837	0.857
	Temporal+Search	25	0.949 [0.87-0.99]	0.864	0.905
	Env_only	46	0.468 [0.32-0.60]	0.568	1.000
	Temporal+Env	53	0.492 [0.33-0.65]	0.568	1.000
	Temporal_only	7	0.263 [0.14-0.40]	0.568	1.000
Respiratory	Full	71	0.936 [0.86-0.99]	0.875	0.966
	Temporal+Search	25	0.938 [0.87-0.99]	0.900	0.931
	Env_only	46	0.788 [0.65-0.90]	0.707	1.000
	Temporal+Env	53	0.778 [0.64-0.89]	0.707	1.000
	Temporal_only	7	0.774 [0.64-0.89]	0.744	1.000

Two patterns emerge. First, predictive discrimination is carried overwhelmingly by the autocorrelated structure of the search series itself: a configuration using only temporal encodings and search-history features (Temporal+Search) matches or slightly exceeds the full model on ROC-AUC for every category (e.g., allergy 0.976 versus 0.957; heat-related 0.949 versus 0.924). Adding the ERA5/CAMS environmental block on top of these features changes ROC-AUC by between −0.002 and −0.025 (Table 12), indicating no measurable incremental discrimination at the national, weekly scale.

Table 11
Trivial baselines on the held-out test set (95% bootstrap CIs). "Last-year-same-week" predicts the binary label observed 52 weeks prior; "AR(search)" is logistic regression on search-interest value lags only.

Category	Baseline	ROC-AUC [95% CI]	F1
Air Quality	Last-year-same-week	0.889 [0.79-0.97]	0.873
	AR(search), LogReg	0.885 [0.79-0.97]	0.833
Allergy	Last-year-same-week	0.365 [0.19-0.56]	0.200
	AR(search), LogReg	0.434 [0.25-0.61]	0.394
Heat-Related	Last-year-same-week	0.382 [0.23-0.55]	0.114
	AR(search), LogReg	0.534 [0.38-0.67]	0.568
Respiratory	Last-year-same-week	0.814 [0.69-0.92]	0.780
	AR(search), LogReg	0.770 [0.62-0.89]	0.710

Table 12
Value-add decomposition (ROC-AUC). "Env value-add" is the change from adding the environmental block on top of Temporal+Search; "Search value-add" is the change from adding search-history on top of Temporal+Env.

Category	Full	Env value-add	Search value-add
Air Quality	0.947	−0.011	+0.046
Allergy	0.957	−0.019	+0.495
Heat-Related	0.924	−0.025	+0.432
Respiratory	0.936	−0.002	+0.158

Second, the environmental data are not uninformative. Used in isolation (Env_only), they retain substantial standalone discrimination for the two biologically plausible categories, respiratory (AUC 0.788) and air quality (0.912), while collapsing to near-random for allergy (0.432) and heat-related illness (0.468). This contrast is itself a result rather than a shortcoming: the env-only collapse for allergy and heat mirrors the weak biological plausibility documented in the Discussion (heat-related search-temperature correlation $r = 0.13$), and is consistent with national aggregation attenuating the localized environmental gradients on which heat exposure and pollen distribution depend. In other words, where the environmental signal is biologically meaningful it is also independently predictive; where it is not, the model recovers its performance entirely from the digital signal's own seasonality and autocorrelation.

Regarding the respiratory model's high recall (0.966), removing all search-history features (Temporal+Env) did not reduce recall; it rose to 1.000, but only by degrading discrimination (AUC from 0.936 to 0.778) and precision. The autocorrelation-free model thus attains maximal recall by predicting the positive class indiscriminately rather than

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by detecting genuine high-risk weeks. The full model's strong sensitivity is therefore not a trivial by-product of search autocorrelation; the search-history features are what preserve discrimination while keeping recall high.

The naive baselines (Table 11) reinforce this reading. Last-year-same-week is competitive for the two categories with stable annual structure (air quality AUC 0.889; respiratory 0.814) but near-random for the seasonally thresholded categories (allergy 0.365; heat-related 0.382), where the within-month percentile labeling breaks the simple year-over-year correspondence. The full pipeline's advantage over the naive predictor is consequently modest for air quality and respiratory and large for allergy and heat-related illness. Across all categories, the pipeline's added value lies less in raw AUC over the naive predictor than in calibrated probabilities (Figure 7) and temporal stability (Table 9) required for risk-based operational use.

4. Discussion

This study developed and rigorously evaluated a machine learning framework for predicting environmental health-related search patterns in Morocco using exclusively freely available global datasets. The results demonstrate that ensemble methods (Gradient Boosting, Random Forest, XGBoost) achieve strong technical performance (ROC-AUC: 0.92-0.96) when applied to satellite-derived environmental data correlated with digital epidemiology signals. Critically, however, biological plausibility varied substantially across the four health categories, underscoring that technical accuracy and epidemiological validity are independent properties requiring separate assessment [17, 20].

4.1. Temporal Feature Dominance

Seasonal anomaly features emerged as by far the strongest predictors across all models under permutation importance (mean AUC drop 0.615 for allergies and 0.228 for respiratory conditions), consistent with epidemiological and climate science findings in European contexts [76]. A key question is whether this dominance primarily reflects underlying seasonal health dynamics or the model's reliance on cyclical encodings. The seasonal anomaly feature captures deviations from the expected week-of-year baseline computed exclusively from training data; unlike simple calendar encodings, it measures *unexpected* search interest relative to seasonal norms, suggesting the models identify genuinely anomalous health-seeking periods rather than merely memorizing calendar position. However, the strong contribution of rolling means and lagged class labels indicates that the models also leverage temporal autocorrelation in search behavior, which may capture habitual search patterns alongside genuine environmental health responses. This dual interpretation has practical implications: the models are well-suited for detecting seasonal and near-seasonal health patterns with high reliability, but may show reduced sensitivity to truly novel, non-seasonal events (e.g., industrial accidents or unexpected pollution episodes) that do not follow historical temporal structures. This interpretation is directly corroborated by the ablation analysis (Section 3.12), which shows that the search-history block accounts for almost all of the discriminative performance.

4.2. Environmental Predictors

Among the environmental features (which rank below the dominant seasonal anomaly), lagged particulate matter terms were the most informative for respiratory outcomes, with $PM_{2.5}$ at a 4-week lag appearing among the contributors. This temporal structure aligns with distributed lag models documented in US studies [77] and Italian research [43], where cumulative inflammatory effects manifest over 2-6 week periods, consistent with hygroscopic particle growth mechanisms documented in global burden studies [50]. The substantial performance gap between ensemble methods (mean AUC: 0.92-0.93) and Logistic Regression (AUC: 0.73) further confirms that these search-environment relationships exhibit complex nonlinear structures, consistent with findings from air quality prediction studies in China [27] and climate-health modeling in the Americas [15]. We note, however, that the ablation analysis shows these environmental features add no incremental discrimination once search-history features are present, and that their value is best understood as standalone, biologically interpretable signal rather than as a source of additional predictive lift (Section 3.12).

4.3. Biological Plausibility Assessment

A critical distinction in digital epidemiology is that high technical performance does not guarantee biological validity. We therefore assess each category against established epidemiological expectations, classifying alignment into three tiers.

Strong alignment - Respiratory conditions. The respiratory model represents the strongest validation case. Elevated predictions concentrated during fall and winter months (October through March), matching typical cold-season respiratory patterns documented in Mediterranean climates [9] and North African contexts [6]. Among the environmental contributors (which are secondary to the seasonal anomaly), lagged particulate matter features such as the 4-week $PM_{2.5}$ lag appear, directionally consistent with distributed lag models documenting cumulative inflammatory effects over 2-6 week exposure windows. The model's high recall (0.966) on the approximately balanced test set (54.7% positive), with only a single borderline false negative, combined with biologically interpretable feature importance patterns, supports potential utility as a respiratory disease surveillance tool.

Moderate alignment - Air quality and allergies. The air quality model showed partial correspondence with expected patterns, including clusters associated with Saharan dust intrusions and winter heating emissions, though the moderate correlation ($r = 0.41$) suggests search behavior is influenced by visibility-based perception alongside actual exposure levels. The allergy model achieved the highest technical performance (AUC: 0.957) but exhibited a summer concentration (June-September) that contrasts with the

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bimodal spring-fall distribution typical of pollen-driven allergies in Mediterranean climates documented in Greek [78] and Spanish studies. The absence of ground-truth pollen monitoring in Morocco precludes definitive interpretation, though the pattern may reflect dust-related mucosal irritation rather than classical pollen allergies [6].

Weak alignment - Heat-related illness. The heat-related model presents fundamental validity concerns. The correlation between search interest and maximum temperature was remarkably weak ($r = 0.13$), with 33.3% of high-search weeks occurring during winter months when temperatures averaged only 18.7°C. Notably, the derived heat index did not rank among the top predictive features for this category, which is itself diagnostic: in a biologically coherent heat-illness signal one would expect it near the top, and its absence is consistent with the weak temperature coupling reported here. This weak coupling may also be partly an artifact of national aggregation: heat exposure is highly local (urban heat islands, coastal versus inland gradients), so averaging maximum temperature across Morocco's diverse geography can dilute precisely the local extremes that would drive heat-related searches. The near-random env-only performance for this category (Section 3.12) is consistent with this interpretation, suggesting that national aggregation contributes to, rather than merely coexisting with, the heat model's weak biological plausibility. This pattern diverges sharply from heat-health studies in France [8] and Spain [79], where search-temperature correlations typically exceed 0.6 during summer months. From an operational perspective, the winter concentration of high-search periods could generate false alerts during periods of minimal actual health risk while potentially missing genuine summer emergencies. We therefore caution against deploying this model for public health alerting without further clinical validation restricted to summer months.

4.4. Data Representativeness and Potential Biases

The representativeness of Google Trends data in Morocco warrants careful consideration. According to the Agence Nationale de Réglementation des Télécommunications (ANRT) annual report for 2024 [80], Morocco's internet penetration exceeds 90% nationally, driven primarily by mobile broadband adoption. However, this aggregate figure masks significant urban-rural disparities: urban centers such as Casablanca, Rabat, and Marrakech exhibit near-universal internet access, while rural areas, particularly in the Atlas Mountain regions and southern provinces, report lower connectivity rates estimated at 40-50%. This geographic imbalance is consistent with broader trends in the MENA region where digital infrastructure remains concentrated in high-density economic hubs [81].

Since Google Trends search volumes are inherently weighted toward connected populations, our models are more representative of urban health-seeking behavior than rural health patterns, a known limitation in digital epidemiology where data "blueprints" favor the digitally visible [82]. This urban bias is, however, consistent with the study's urban environmental health framing, which targets populations in urban environments where both environmental health risks and digital infrastructure are concentrated. Nevertheless, caution is warranted when extrapolating these findings to rural populations or interpreting national-level predictions as uniformly representative across all regions.

Beyond internet penetration, three additional sources of bias may affect the validity of search-based health signals. First, *media amplification* can generate search spikes driven by news coverage of health events occurring elsewhere, unrelated to local health burden [83]; this effect is particularly relevant for the heat-related category, where Moroccan search interest may reflect coverage of European heatwaves rather than domestic heat exposure. Second, *awareness campaigns* conducted by government health authorities can artificially elevate search volumes for specific health terms, creating confounders that models cannot distinguish from genuine symptomatic health-seeking behavior. Third, *algorithmic content recommendations* on search platforms may create feedback loops where trending topics generate further searches independent of actual health conditions, a phenomenon described as "algorithmic confounding" [17]. While these biases affect all digital epidemiology approaches, their potential influence should be considered when interpreting model predictions, particularly for categories where biological plausibility is already weak.

4.5. Correlational Nature and Causal Inference

An important methodological consideration is that the predictive associations identified in this study are correlational rather than causal. The observed high predictive accuracy should be interpreted primarily as the models' ability to capture repeated temporal and behavioral patterns in information-seeking, rather than as evidence of direct causal relationships between environmental exposures and health outcomes. Establishing causality would require experimental or quasi-experimental designs, such as Granger causality testing to assess temporal precedence between environmental changes and search behavior shifts, or distributed lag nonlinear models (DLNMs) that can simultaneously model the nonlinear and delayed effects of environmental exposures [34]. The ecological study design further limits causal interpretation, as population-level associations may not reflect individual-level exposure-outcome relationships (ecological fallacy). Future extensions of this work should prioritize these causal inference methodologies to distinguish genuine health-seeking responses from confounded associations.

4.6. Operational Deployment and Implementation Strategy

The operational feasibility of real-time deployment depends on data latency, computational requirements, and system integration. ERA5 reanalysis data are available with approximately 5-day delay through the Copernicus Climate Data Store, CAMS near-real-time forecasts within 24 hours, and Google Trends at weekly resolution with a 2-3 day processing lag. These latencies collectively enable near-weekly

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prediction updates suitable for early warning at weekly time scales. Model inference requires under 1 second on standard hardware (Intel i7, 16 GB RAM), and the full cycle from data retrieval through feature computation to inference can be executed within minutes using scheduled scripts.

Even in the absence of clinical validation, the framework offers practical value as a low-cost screening and resource-allocation layer rather than a standalone surveillance system. In settings where traditional syndromic surveillance suffers from delayed reporting and incomplete coverage, search-based predictions can serve as an initial filter that triggers enhanced monitoring or targeted epidemiological investigation. This screening function is most relevant for the respiratory category, whose predictions align with expected disease dynamics and could prompt pre-positioning of hospital resources during anticipated high-burden weeks.

These properties suggest a phased strategy. In the short term (2025-2027), the respiratory model should be prioritized for pilot deployment in shadow mode alongside existing surveillance, with partnerships focused on validating its digital signals against hospital admission records to establish the epidemiological grounding that technical metrics alone cannot guarantee [9, 17]. Concurrently, the instability of heat-related search patterns underscores the need to monitor feature drift, as digital signal validity may shift over time independent of climate-health relationships [18]. In the medium term (2027-2030), efforts should shift toward integrated early warning systems combining satellite-derived environmental data, validated search signals, and clinical records rather than any single source [28], supported by stronger local air quality monitoring to validate satellite estimates [25], capacity building in digital epidemiology interpretation [19], public-private data partnerships, and regional collaboration across North Africa to test transferability. Throughout, the framework should be positioned as complementary decision support pending external validation, not as a replacement for clinical surveillance.

4.7. Generalizability Considerations

While the findings from Morocco provide relevant insights for other resource-constrained settings in the Global South, it is important to recognize that the country also presents unique features due to its diverse climate zones and specific exposure profile combining Saharan dust with anthropogenic pollution [6]. National-level aggregation may differentially affect each health category: heat-related signals are more sensitive to local urban heat island effects than respiratory patterns driven by large-scale seasonal dynamics. These characteristics may limit the direct generalizability of category-specific validity findings to countries with different climatic or pollution contexts.

5. Summary and Perspectives

This study delivers a comprehensive, data-driven analysis of environmental health prediction in Morocco using exclusively freely available global datasets. By integrating five years of weekly data (April 2020-April 2025) from ERA5 meteorological reanalysis, CAMS atmospheric products, and Google Trends, we developed machine learning models for four search categories used as digital proxies: air quality concerns, respiratory conditions, heat-related illness, and allergies. The methodology incorporated extensive feature engineering (71 features including temporal encodings, environmental lags, interaction terms, and seasonal anomalies), rigorous time series cross-validation, probability calibration, and expanding-window backtesting to ensure robust temporal validation.

Using ensemble methods including Gradient Boosting, Random Forest, and XGBoost, we achieved strong technical performance across all categories (ROC-AUC: 0.92-0.96; F1-scores: 0.84-0.92) with temporal stability over 52-week forecasting horizons (see Tables 8 and 9). Feature importance analysis identified the seasonal anomaly as the dominant predictor across categories, with lagged particulate matter and temperature-humidity interactions as secondary environmental contributors. An ablation analysis (Section 3.12) further showed that this discrimination is carried primarily by the autocorrelated search signal, while the environmental block contributes interpretable standalone signal for the biologically plausible categories but limited incremental value at national scale. Critically, our tiered biological plausibility assessment revealed that the respiratory model demonstrated strong alignment with established epidemiological patterns, while the heat-related illness model exhibited fundamental validity concerns requiring clinical validation before operational use.

This work fills a pivotal gap in environmental health surveillance for North Africa by demonstrating that satellite-derived environmental data combined with digital epidemiology signals can support health monitoring in resource-constrained settings, contributing to SDG 3, 11, and 13 implementation. By focusing on Morocco, this study offers a replicable framework potentially transferable across the Global South, where traditional surveillance infrastructure remains limited. The finding that technical accuracy and biological validity are independent properties requiring separate evaluation provides essential methodological guidance for digital epidemiology research.

Nevertheless, some limitations remain. The use of national-level aggregated data cannot fully capture local extremes in urban heat islands or industrial areas. The ecological fallacy inherent in population-level predictions prevents interpretation as individual-level causal effects. Additionally, the absence of clinical outcome data (emergency department visits, hospital admissions, mortality) restricts definitive validation of search-disease concordance. This is not merely a future refinement but a present boundary on interpretation: until external validation against hospital admissions, emergency visits, or mortality is completed, these models cannot be interpreted as validated predictors of disease burden, and this caveat applies with particular force to the heat-related model, whose weak temperature alignment and implausible winter peaks are documented above. Allergy

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model validation is further limited by the lack of ground-truth pollen monitoring in Morocco.

To address these limitations, future work will focus on several directions.

First, clinical validation is essential before operational deployment. We aim to establish partnerships with Moroccan health authorities, specifically the Ministry of Health Direction of Epidemiology, to access anonymized hospital admission and emergency department records, enabling direct comparison between search-based predictions and actual health outcomes. For the respiratory model, validation will assess whether the 4-week particulate matter lag structure corresponds to clinical presentation timing. For heat-related illness, validation will focus exclusively on summer months to test whether winter search peaks correspond to any measurable health burden.

Secondly, mechanistic investigation of category-specific search drivers will help distinguish genuine health-seeking behavior from the media-driven, campaign-related, and algorithmic biases identified in the Discussion. For heat-related queries, targeted studies should quantify the relative contribution of each bias source to observed search patterns. For allergy searches, concurrent pollen monitoring would clarify whether summer concentration reflects dust exposure or other non-pollen triggers.

Third, causal inference methodologies including Granger causality testing and distributed lag non-linear models (DLNMs) should be applied to investigate the directionality and temporal structure of environment-search relationships beyond the correlational framework of the current study.

Finally, spatial disaggregation to the city and regional level represents the priority extension of this work. The present national-scale prototype is constrained by the sub-national sparsity of Google Trends documented above; future work will therefore target city-level modeling by integrating denser sub-national signals, such as social media activity, telecom mobility data, and local air-quality sensor networks, alongside or in place of national search aggregates. High-resolution analysis at the city scale may also reveal whether local environmental extremes (urban heat islands, localized industrial plumes) produce more biologically plausible search-environment relationships than the national aggregates used here, particularly for the heat-related category whose weak coupling we attribute partly to aggregation. This effort will support the development of localized early warning systems for Moroccan cities, combining predictive models with real-time environmental monitoring to assist public health authorities in evidence-based decision-making for climate-sensitive health risks.

CRediT authorship contribution statement

Ayoub Lahlouh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing (original draft), Visualization. **Jérôme Chenal**: Supervision, Writing (review & editing). **Naoufal El Hachlafi**: Methodology, Validation, Writing (review & editing). **Mariem Bounabi**: Methodology, Validation, Writing (review & editing). **Mohammed Dalli**: Methodology, Validation, Writing (review & editing). **Mohammed Hlal**: Investigation, Data curation, Writing (review & editing). **Seyid Abdellahi Ebnou Abdem**: Methodology, Validation, Writing (review & editing). **Rida Azmi**: Supervision, Writing (review & editing). All authors have read and agreed to the published version of the manuscript.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Anthropic's Claude in order to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Data Availability Statement

ERA5 reanalysis data are available from the Copernicus Climate Data Store (https://cds.climate.copernicus.eu/). CAMS air quality data are available from the Copernicus Atmosphere Data Store (https://ads.atmosphere.copernicus.eu/). Google Trends data can be accessed at https://trends.google.com/. All code for data processing, feature engineering, model training, ablation, and evaluation, together with the pinned environment specification, is publicly available at `https://github.com/ayoubgeo/urban-health-morocco` and permanently archived at Zenodo (DOI: `https://doi.org/10.5281/zenodo.20476258`). The repository includes the model predictions, permutation-importance rankings, and evaluation metrics reported here.

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Ethics Statement

This study used publicly available aggregated search data and environmental monitoring data. No individual-level data or personally identifiable information were collected or analyzed. Ethical approval was not required as the study did not involve human subjects research.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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